MedSynth: Realistic, Synthetic Medical Dialogue-Note Pairs

Abstract

Physicians spend significant time documenting clinical encounters, a burden that contributes to professional burnout. To address this, robust automation tools for medical documentation are crucial. We introduce MedSynth – a novel dataset of synthetic medical dialogues and notes designed to advance the Dialogue-to-Note (Dial-2-Note) and Note-to-Dialogue (Note-2-Dial) tasks. Informed by an extensive analysis of disease distributions, this dataset includes over 10,000 dialogue-note pairs covering over 2,000 ICD-10 codes. We demonstrate that our dataset markedly enhances the performance of models in generating medical notes from dialogues, and dialogues from medical notes. The dataset provides a valuable resource in a field where open-access, privacy-compliant, and diverse training data are scarce.

Keywords: medical text, medical dialogue summarization, synthetic data, benchmark

Data and Code Availability In this study, we use three distinct data sources. The IQVIA PharMetrics Plus database was used to estimate the distribution of ICD-10 [1] codes and is accessible by obtaining a license by contacting IQVIA company[2]. Additionally, we use Aci-Bench data (Yim et al., 2023b), which appears to be the largest relevant, publicly available dataset in terms of the length of dialogues and notes that went through extensive medical expert evaluation. We incorporate this dataset's training set in our data generation pipeline for In-Context Learning (ICL), and its test set to evaluate the effectiveness of our synthetic data generation pipeline. We also use NoteChat (Wang et al., 2023). It contains medical notes and synthetic dialogues and, to our knowledge, is the largest dataset in terms of the number of data points. However, their notes are patient summaries and do not follow the structure of standard medical notes. We use this dataset in our empirical evaluation. Our synthetic dataset and code are publicly available at [URL redacted for anonymity].

Institutional Review Board (IRB) This research does not require IRB approval.

1. Introduction

Electronic health records (EHRs) are widely used and improved clinical documentation, enabling more streamlined patient care (Melton et al., 2021). However, despite their numerous benefits, EHRs also introduce a significant documentation burden on healthcare providers. Physicians report spending between 52 to 102 minutes daily on clinical note-taking from patient encounters, a task that is both time-consuming and detracting from direct patient care (Hripcsak et al., 2011). This extensive documentation requirement contributes significantly to physician burnout, highlighting an urgent need for solutions that can mitigate these demands while preserving the quality and integrity of patient records.

Automated tools for medical documentation have emerged as promising solutions to reduce this burden. However, developing and validating these tools is often hampered by the lack of large, open-access datasets that are both comprehensive and privacy-compliant. Existing datasets tend to be limited either in scope, covering only a small number of medical conditions, or by not being publicly available due to privacy concerns, thus restricting the potential for widespread adoption (Wang et al., 2023; Yim et al., 2023b). Table 1 shows a comparison of the available datasets. The latest openly accessible benchmark for the Dial-2-Note task is Aci-Bench, which contains only 207 data points. With 207k data points, NoteChat appears to be the largest relevant dataset, containing medical notes and synthetic dialogues but is limited to cancer-related cases (Zhao et al., 2022), and the notes do not follow medical standards in structure but, rather, are patient summaries.

We address the limitations of existing datasets by providing a novel synthetic dataset that serves both as an augmentation strategy for training and a benchmark for future model developments. This dataset, being fully synthetic, circumvents the prevalent pri-

1. International Classification of Diseases, 10^{th} revision
2. https://www.iqvia.com/

Table 1: Comparison of Dial-2-Note Generation Datasets

Dataset	**Description**	**Src-len (tok/turns)**	**Target-len (tok/sent)**	**Size**	**Open**
3M Health (Zhang et al., 2021)	Dialogue-note pairs: notes are created using dialogues	-/-	-/- (HPI only)	1,342	N
Abridge (Krishna et al., 2020)	Dialogue-note pairs: notes are created using dialogues	1500/-	-/27	6,862	N
Augmedix (Yim and Yetisgen-Yildiz, 2021)	Real clinical dialogue-note pairs	-/175	-/47	500	N
emr.ai (Finley et al., 2018)	Real clinical dialogue-note pairs	616/1	550/-	9,875	N
Nuance (Enarvi et al., 2020)	Real clinical dialogue-note pairs	972/-	452/-	802k	N
MTS-dialogue (Abacha et al., 2023b)	Dialogue-note snippets: dialogues are created using clinical note sections	142/9	48/3	1,701	Y
primock57 (Korfiatis et al., 2022)	Dialogue-note pairs: role-played by human	1,489/97	161/23	57	Y
Aci-Bench (Yim et al., 2023b)	Dialogue-note pairs: role-played by human	1,302/55	490/49	207	Y
NoteChat (Wang et al., 2023)	Dialogue-note pairs: dialogue role-played by LLMs based on patient summaries	561/27	301/14	207k	Y
MedSynth (this work)	Dialogue-note pairs: notes and dialogues role-played by LLMs	1,070/47	620/23	10,035	Y

vacy issues associated with using real patient data. Our primary contributions are:

1. **A large synthetic medical dialogue-note dataset:** We provide a comprehensive, privacy-compliant dataset of medical dialogues paired with clinical notes encompassing a wide array of medical conditions over more than 2000 different ICD-10 codes and over 10,000 dialogue-note pairs. The notes in our dataset follow the SOAP structure (Subjective, Objective, Assessment, Plan), which is one of the most common formats of medical notes (Vivek Podder and Valerie Lew and Sassan Ghassemzadeh, 2024). The diversity of the ICD-10 codes and the use of SOAP structure in our study are mainly to mimic the use case of primary care. However, our data contains diseases related to other specialties as well. We introduce a new benchmark for dialogue-to-note generation tasks using the dataset.
2. **Data generation pipelines:** We introduce a novel pipeline for generating synthetic medical dialogue-note pairs, designed to assist researchers and developers in creating synthetic medical notes and dialogues.
3. **SOTA Dia-2-Note generation models:** We release models fine-tuned on our synthetic dataset[3] that achieve SOTA performance in generating SOAP medical notes from doctor-patient dialogues and can be used as the base model for developing tools to automate documentation.

2. Related Work

In medical dialogue-to-note summarization, several datasets are commonly used, including 3M Health (Zhang et al., 2021), Abridge (Krishna et al., 2020), Augmedix (Yim and Yetisgen-Yildiz, 2021), emr.ai (Finley et al., 2018), and Nuance (Enarvi et al., 2020)

3. [URL redacted for anonymity]

are important tools for developing and testing models. However, their lack of public availability restricts their utility. To address this challenge, efforts have led to the development of open-source datasets such as MTS-dialogue (Abacha et al., 2023b), primock57 (Korfiatis et al., 2022), Aci-Bench (Yim et al., 2023b), and NoteChat (Wang et al., 2023). These datasets represent a meaningful advancement in improving the accessibility of medical dialogue data. Notably, some of these datasets, such as Aci-Bench, include synthetic dialogues created through human role-playing, while others, like NoteChat (Wang et al., 2023), use large language models (LLMs) to generate dialogues. Despite these advancements, existing datasets still have various limitations. They often concentrate on specific diseases, lacking a comprehensive range of medical conditions common in primary care, and none of them address the simultaneous synthesis of both medical notes and dialogues.

3. Methodology

We introduce the data sources used, analyze the disease distribution, and discuss important quality factors identified through a comprehensive survey, which informs our dataset construction. The subsequent parts of this section describe the pipelines developed for generating medical notes and dialogues.

3.1. Data Resource

Three main data sources inform this research and are introduced in this section.

3.1.1. IQVIA PharMetrics Plus

IQVIA PharMetrics Plus is a US medical insurance claims database, which has been the data source for several important research findings such as Yaghmaei et al. (2023) and Zheng et al. (2024). We had access to a 25% random sample of the database from 2006 to 2021, which amounts to over 3 billion insurance claims.

The database contains claims with either ICD-9 or ICD-10 code labels. ICD-10 codes were introduced and applied in 2015. It is challenging to map ICD-9 codes to ICD-10 because ICD-10 is more granular and contains more codes than iCD-9 (Cartwright (2013)). Therefore, we filtered the data to only include claims with ICD-10 codes. According to (Benes, 2023), some codes are duplicated across the ICD-9 and ICD-10 systems. To avoid confusion, we removed the duplicates and ended up with 800 million claim records and 54,985 unique ICD-10 codes.

3.1.2. Aci-Bench

Yim et al. (2023b) introduced the Aci-Bench dataset, which contains 207 dialogue-note pairs. The data are generated by medical experts and lay persons role-playing. To the best of our knowledge, the dataset is the largest publicly available dataset in terms of the length of the dialogues and notes that went through extensive human evaluation.

3.1.3. NoteChat

This dataset is introduced by Wang et al. (2023) and contains 207k dialogue-note pairs. The notes are patient summaries extracted from PubMed, and the dialogues are synthetically generated using the cooperation of multiple role-playing LLMs. NoteChat is one of the largest publicly available datasets in this area in terms of the number of data points. However, their data are mostly (proportion unspecified) related to cancer cases, and the notes are not provided in standard medical formats but are patient summaries (Zhao et al., 2022).

3.2. Diseases Distribution Analysis

We start by identifying the distribution of the most prevalent diseases. To do so, we use the IQVIA PharMetrics Plus database introduced in Section 3.1.

Table 2: Distribution of ICD-10 codes from IQVIA Pharmetrics Plus database.

# of Claims	# of ICD-10 Codes
Above 10M	4
1M - 10M	112
100k - 1M	1032
10k - 100k	3922
1k - 10k	8942
Below 1k	40,973

Table 2 shows that the distribution is highly skewed. We choose the top 2,000 ICD-10 codes and generate 5 dialogue-note pairs for each. Our goal is to generate a synthetic dataset that can be used to train models that can handle more diverse diseases than is currently supported; therefore, we sample uniformly from the distribution of ICD-10 codes. To the best of our knowledge, our dataset is the largest dataset of its type available publicly in terms of the number and diversity of diseases.

3.3. Survey on Important Quality Factors

As mentioned by Mathioudakis et al. (2016), medical notes should adhere to standards to ensure continuity of care and facilitate communication between healthcare professionals. Good documentation should allow patients to access and understand the treatment procedure. Moreover, in case of an audit of healthcare services, medical notes can be a valuable resource.

To ensure the synthetic medical notes capture useful information, particularly because real medical notes were inaccessible, we surveyed two medical professionals to determine the essential variables for note quality. One participant is a general physician and the other is a bio-statistician with years of experience with EHR systems and medical data. We listed 26 variables and asked the raters to score each variable between 0 to 10 in terms of how important each variable is in determining the quality of synthetic medical notes. We then took the top 10 variables with the highest average importance scores. There was a tie in the 10^{th} position, so we considered 11 variables initially. Figure 24 in Appendix E shows the results of the survey.

Also, after running the survey and discussing it with the medical experts, we added two more variables; Physical Exams and Investigations/Tests.

We use the results from this survey and the disease distribution analysis to guide our data generation pipelines.

3.4. Data Generation Pipeline

In our data generation methodology, we first generate notes and then use the notes to generate the corresponding dialogues. We have multiple LLMs collaborating to generate these notes. We use GPT-4o in both note generation and dialogue generation pipelines. Two prompting strategies were used in our data generation pipelines: Chain-of-Thought (CoT) (Wei et al., 2022) improves LLM performance in complex reasoning tasks through providing intermediate reasoning steps, and role-playing prompting (Kong et al., 2023) that further improves performance. All prompts can be found in Appendix A.

We also use In-Context Learning (ICL) as a training-free framework and that not require updating model parameters (Brown, 2020). It is directly applied to a pre-trained LLM and improves model performance by "showing" the model what a good answer to a query looks like without updating model parameters. We refer to Dong et al. (2022) for an in-depth definition and a survey of ICL. In this paper, we use three forms of ICL:

a) Few-Shot Learning: At inference time, n demonstrations of the desired task are given to the model as conditioning in natural language.

b) One-Shot Learning: It is similar to Few-Shot Learning, with n=1.

c) Zero-Shot Learning: It is similar to Few-Shot Learning, but instead of providing examples of the required task, a natural language description of the task is provided to the model.

3.4.1. Note Generation Pipeline

Figure 1 shows that our note-generation pipeline that consists of four LLMs:

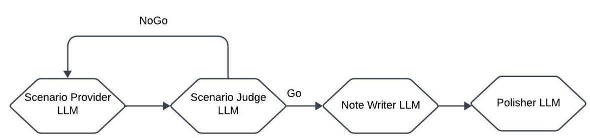

Figure 1: Note Generation Pipeline: The pipeline consists of four LLMs collaborating to generate the medical notes

a) Scenario Provider LLM: This LLM is responsible for generating scenarios given the description of the ICD-10 code. The Scenario Provider receives the disease description (ICD-10 description), selects an appropriate role based on the disease (e.g., if the disease involves a broken bone, the Scenario Provider chooses Orthopedist), and generates a scenario based on the 13 specified variables that fit the disease description and the chosen role. In each generation, we sample a note from the Aci-Bench training set and provide it to the model.

b) Scenario Judge LLM: The Scenario Judge LLM is tasked with evaluating the quality and validity of scenarios generated by the Scenario Provider LLM. Its main responsibility is to determine whether the provided scenarios meet specific conditions to be approved for generating synthetic medical notes. The evaluation process involves three checks:

i) Comparison of Variables: The LLM checks whether at least four out of the thirteen variables in the current scenario differ from previously approved scenarios for the same ICD-10 code. This ensures diversity and uniqueness in the generated scenarios.

ii) Medical Accuracy: The LLM examines the scenario to confirm that it is medically accurate. This includes verifying that the symptoms, tests, diagnosis, and treatment are appropriate and logically consistent with the disease described by the ICD-10 code description.

iii) Plausibility: The LLM evaluates the overall plausibility of the scenario to ensure it is realistic and could feasibly occur in a medical setting.

If a scenario is rejected, detailed feedback will be provided by the Scenario Judge to the Scenario Provide. The feedback contains textual suggestions from the Scenario Judge about what parts of the scenario need to be improved to pass the checks. This loop continues until a pre-selected number of scenarios are approved.

c) Note Writer LLM: The Note Writer LLM receives a scenario detailing the scenario and physician's role, which could be a Family Medicine Physician, a General Physician, or a specialist in various fields. Appendix C shows the statistics of the roles. Based on this scenario and the role, the LLM generates a medical note adhering to the SOAP format:

Subjective: The patient's description of their symptoms and complaints.

Objective: The physician's observations and collected data, including vital signs, physical examination findings, and test results.

Assessment: The physician's evaluation of the patient's condition, including the diagnosis or differential diagnosis.

Plan: The physician's recommendations for treatment, management, and follow-up.

The Note Writer LLM uses One-Shot Learning; in each generation, we sample a note from the Aci-Bench training set and provide it to the model to help generate notes in a similar format.

d) Note Polisher LLM: The Note Polisher LLM is designed to refine and enhance synthetic medical notes. The primary task is to ensure that the information in the notes is correctly categorized and placed in the appropriate sections.

All the prompts provided to each LLM in each stage are available in Appendix A.

3.4.2. Dialogue Generation Pipeline

Our data generation pipeline consists of two LLMs: the Dialogue Generator and the Dialogue Polisher.

a) Dialogue Generator LLM: This LLM generates a dialogue relevant to a given medical note. We use Few-Shot Learning with n=3 and feed three examples from the Aci-Bench training set to boost the quality of the generation. The dialogue must include all the information from the medical note and follow specific guidelines to ensure authenticity and comprehensiveness.

b) Dialogue Polisher LLM: The Dialogue Polisher LLM enhances and expands doctor-patient conversations (including social chatter) to make them more realistic and engaging while ensuring all information from the medical notes is accurately included.

4. Results

In this section, we start with a detailed analysis of MedSynth's characteristics. Following this, we indirectly assess the quality of MedSynth by utilizing it in Dial-2-Note and Note-2-Dial tasks, demonstrating its effectiveness and versatility. Finally, we establish a new benchmark based on MedSynth for the Dial-2-Note task, setting a standard for future innovations and assessments in the field of medical documentation automation.

4.1. Data Characteristics

This subsection details the key features and statistics of the MedSynth dataset, highlighting its relevance to medical documentation. To analyze the differences between the MedSynth, Aci-Bench, and real doctor-patient encounter datasets, we performed a statistical comparison across 20 samples. Our analysis could only utilize the 20 samples provided for comparison in the Aci-Bench study, as real encounter data was not directly available to us. In this comparison, the MedSynth dataset typically featured the shortest dialogue lengths, averaging 970 tokens without speaker tokens, compared to 1505 and 1203 tokens for Aci-Bench and real data respectively. However, dialogue sentence length was greatest in Aci-Bench at 141 sentences, with real data and MedSynth following at 80 and 57 sentences respectively. In terms of clinical notes, Aci-Bench notes were the longest, averaging 683 tokens, while MedSynth and real data notes were shorter, with 618 and 492 tokens respectively. Note sentence lengths also varied, with Aci-Bench having the longest notes at 66 sentences, followed by real data at 49 sentences, and MedSynth at 23 sentences. Table 8 in the Appendix F illustrates the detailed comparison.

4.2. Usefulness of the Data for the Dial-2-Note and Note-2-Dial Generation

Human evaluation of generated data is costly and time-consuming (Liu et al. (2024)). Therefore, we take an automatic approach to assessing the quality and usefulness of the data for Dial-2-Note and Note-2-Dial generation.

Aci-Bench represents the latest benchmark for these tasks. The central hypothesis is that if MedSynth's quality is high and its dialogues and notes closely resemble the distribution of Aci-Bench, then models trained on MedSynth should achieve strong performance on the Aci-Bench test set.

4.2.1. Evaluation Metrics

We follow Abacha et al. (2023a), Yim et al. (2023a), and Van Veen et al. (2024) and evaluate with BLEU (Papineni et al., 2002), ROUGE-1, ROUGE-2, ROUGE-L,ROUGE-LSum (Lin, 2004), and METEOR (Banerjee and Lavie, 2005). Current pretrained embedding-based evaluation metrics (such as bertscore, bleurt, and bart-score) are limited by the original trained sequence lengths which are usually much shorter than our full notes and dialogues. Figure 25 in the Appendix E shows the distribution of note lengths by subtoken.

4.2.2. Results

We ran a series of experiments to indirectly evaluate the quality of MedSynth across two tasks: Dial-2-Note and Note-2-Dial. For the Dial-2-Note task, we sampled data from MedSynth and NoteChat in different sizes, combined these with the Aci-Bench training set, and trained the Llama-3-8B-Instruct model. The performance was then tested on the Aci-Bench test set. To further assess the standalone value of MedSynth, we ran an additional experiment where the Aci-Bench training set was excluded from training. We observed that the model trained solely on MedSynth not only outperformed the model trained on NoteChat but also matched or surpassed the performance of the model trained exclusively on the Aci-Bench training set across most metrics.

In the Note-2-Dial task, we ran experiments using the entire MedSynth dataset and a sample from NoteChat with the same size. We trained the Llama-3-8B-Instruct model under various conditions: one where the Aci-Bench training set was included alongside MedSynth and NoteChat, and another where it was excluded. The models were then evaluated on the Aci-Bench test set.

Our analysis of the MedSynth dataset on both the Note-2-Dial and Dial-2-Note tasks shows consistent performance enhancements when integrating MedSynth with the Aci-Bench training set compared to NoteChat. Table 3 shows the results of the Dial-2-Note task and Table 4 shows the results of the Note-2-Dial task. The results indicate that variations in dataset size did not significantly alter the performance outcomes for MedSynth and NoteChat. It suggests that the improved metrics can be attributed to qualitative factors inherent in the datasets rather than merely to their sizes.

An important observation is that when the models were trained exclusively on MedSynth compared to exclusively on NoteChat, a significant disparity in performance was noted in their adherence to the SOAP format. Specifically, the model trained only on NoteChat failed to maintain the structured SOAP format, which is a critical component of standard medical documentation. This confirms our expectation because NoteChat includes patient summaries instead of structured medical notes. An example of such a failure is provided in Appendix D.

4.3. Benchmarking

We split the MedSynth benchmark into a train set of 85% of the data and the remaining 15% in the test set to introduce a new benchmark. The training set includes 8,529 dialogue-note pairs with 2,001 unique ICD-10 codes, while the test set includes 1,506 data points with 1,131 unique ICD-10 codes. Table 9 in the Appendix F shows the detailed statistics of the training and test sets.

For the experiments, GPT-4o was used with Zero-Shot Learning. Llamma-3-8B-Instruct and Mistral-7b-instruct-v0.3, both 4-bit quantized were fine-tuned with Low-Rank Adaptation (LoRA) (Hu et al. (2021)) on the training set for one epoch. The prompts were the same for all the experiments to ensure fair comparison. The prompts can be found in Appendix A and the hyperparameters in the experiments can be found in Appendix B. The results

Table 3: Comparison of model performance by dataset and sample size for Dial-2-Note task.'MS+AT': MedSynth+Aci_Train; 'NC+AT':NoteChat+Aci_Train.

Samples	**Dataset**	**BLEU**	**ROUGE-1**	**ROUGE-2**	**ROUGE-L**	**ROUGE-LSum**	**METEOR**
67	Aci_Train	0.12	0.48	0.23	0.29	0.44	0.36
567	MS+AT	0.29	0.60	0.33	0.38	0.55	0.46
567	NC+AT	0.23	0.57	0.31	0.36	0.53	0.44
1567	MS+AT	0.28	0.60	0.34	0.40	0.56	0.46
1567	NC+AT	0.25	0.58	0.33	0.38	0.54	0.45
2567	MS+AT	0.31	0.61	0.35	0.40	0.57	0.47
2567	NC+AT	0.26	0.59	0.32	0.39	0.55	0.46
4067	MS+AT	0.26	0.58	0.33	0.38	0.54	0.45
4067	NC+AT	0.27	0.60	0.34	0.40	0.56	0.47
5567	MS+AT	0.28	0.60	0.34	0.41	0.56	0.45
5567	NC+AT	0.28	0.60	0.34	0.40	0.56	0.46
7067	MS+AT	0.31	0.62	0.35	0.41	0.57	0.46
7067	NC+AT	0.26	0.60	0.33	0.39	0.56	0.48
8567	MS+AT	0.29	0.61	0.34	0.41	0.57	0.45
8567	NC+AT	0.26	0.58	0.32	0.37	0.54	0.44
10102	MS+AT	0.31	0.62	0.35	0.41	0.57	0.47
10102	NC+AT	0.28	0.61	0.34	0.39	0.56	0.47
10035	MedSynth	0.12	0.52	0.23	0.30	0.48	0.38
10035	NoteChat	0.07	0.38	0.15	0.22	0.33	0.34

Table 4: Comparison of model performance by dataset and sample size for Note-2-Dial task. 'MS+AT': MedSynth+Aci_Train; 'NC+AT':NoteChat+Aci_Train.

Samples	**Dataset**	**BLEU**	**ROUGE-1**	**ROUGE-2**	**ROUGE-L**	**ROUGE-LSum**	**METEOR**
67	Aci_Train	0.09	0.37	0.14	0.18	0.34	0.26
10102	MS+AT	0.15	0.54	0.24	0.25	0.51	0.33
10102	NC+AT	0.13	0.51	0.24	0.25	0.49	0.29
10035	MedSynth	0.08	0.51	0.19	0.23	0.49	0.29
10035	NoteChat	0.03	0.43	0.17	0.22	0.41	0.19

show that Mistral-7b-instruct-v0.3 outperforms the other models. Another important observation is that, fine-tuned smaller specialized models outperform the larger generalist model on medical Dial-2-Note tasks in our settings. The results of the experiments with three models on the new benchmark are presented in Table 5.

5. Discussion

We introduce MedSynth, a novel synthetic dataset designed to advance dialogue-to-note generation, and to establish a new benchmark for evaluating model performance in this domain. To the best of our knowledge, MedSynth is the first dataset of fully synthetic medical notes and dialogues, a crucial innova-

Table 5: Benchmarking models on MedSynth: GPT-4o is used in zero-shot, and Llama and Mistral models are trained on the train set and tested on the test set

Model	BLEU	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-LSum	METEOR
unsloth/mistral-7b-instruct-v0.3	0.53	0.74	0.51	0.59	0.72	0.66
unsloth/llama-3-8b-Instruct-bnb-4bit	0.52	0.73	0.50	0.57	0.71	0.65
GPT-4o (Zero-Shot)	0.26	0.65	0.40	0.48	0.63	0.48

tion in a field where real data is scarce due to stringent privacy concerns. By incorporating over 2000 ICD-10 codes and more than 10,000 dialogue-note pairs, MedSynth overcomes the absence of available, diverse, and privacy-compliant datasets for medical text generation. Although the focus of MedSynth is on primary care, where observed diseases are vast, it also contains diseases related to other specialties and can be used in any context that follows the SOAP format.

We show that models trained on MedSynth exhibit improvements in generating accurate and contextually appropriate medical notes from dialogues, which is essential for reducing the documentation burden on healthcare providers and addressing physician burnout.

Another contribution of this work is the creation of a new benchmark against which future models can be evaluated. Existing datasets, such as Aci-Bench and NoteChat, though valuable, are drastically limited either in the number of data points, the diversity of medical conditions covered, or the standard structure of medical notes. Aci-Bench, for instance, is comprehensive in its evaluation by medical experts but limited in size with only 207 dialogue-note pairs. NoteChat, on the other hand, provides a larger dataset with 207k data points but uses patient summaries rather than structured medical notes and is mostly limited to cancer.

MedSynth addresses these limitations by offering a dataset that is not only more diverse but also adheres to medical structures typically found in clinical documentation, especially common in primary care.

Finally, we release our fine-tuned models that achieve SOTA performance in the targeted tasks, which can be used as base models for future tool development.

6. Limitations and Ethical Considerations

While MedSynth provides a substantial resource and step forward, there are some limitations to consider. First, while our pipeline is effective in generating synthetic data to improve Dial-2-Note and Note-2-Dial models, it does not guarantee the medical correctness of the generated data. Although initial experiments do not show cause for concern, further research and expert evaluation is necessary to ensure medical accuracy. Therefore, MedSynth should not be considered a source of reliable medical information but rather a tool for model development.

Second, the synthetic nature of MedSynth, though beneficial for privacy, may not fully capture the complexities of real-world clinical dialogues between humans. Future work should evaluate the realism of these dialogues and expand the dataset to include a broader range of diseases beyond the top 2,000 ICD-10 codes, increasing its overall utility.

Third, although SOAP is among the most widely used structures for recording medical notes, other structures exist. We focused on SOAP because it is among the most common structures in primary care. Future work should consider covering more structures of medical notes.

Finally, while our results improve model performance in controlled settings, it is crucial to practice safe, careful 'MLOps' techniques when evaluating these models in clinical environments. Validating MedSynth's impact on documentation efficiency and accuracy in real-world clinical workflows, with 'real world' indicators of performance, will be an essential next step.

Acknowledgments

[Acknowledgements redacted for anonymity]

References

Asma Ben Abacha, Wen-wai Yim, Griffin Adams, Neal Snider, and Meliha Yetisgen-Yildiz. Overview of the mediqa-chat 2023 shared tasks on the summarization & generation of doctor-patient conversations. In *Proceedings of the 5th Clinical Natural Language Processing Workshop*, pages 503–513, 2023a.

Asma Ben Abacha, Wen-wai Yim, Yadan Fan, and Thomas Lin. An empirical study of clinical note generation from doctor-patient encounters. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2291–2302, 2023b.

Satanjeev Banerjee and Alon Lavie. METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In *Proceedings of the acl workshop on intrinsic and extrinsic evaluation measures for machine translation and/or summarization*, pages 65–72, 2005.

Gregory Benes. Risk of duplicate ICD codes for orthopedic and injury related research. *Perspectives in health information management*, 20(1), 2023.

Tom B Brown. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*, 2020.

Donna J. Cartwright. ICD-9-CM to ICD-10-CM Codes: What? Why? How? *Advances in Wound Care*, 2(10):588–592, 2013. doi: 10.1089/wound.2013.0478. URL https://doi.org/10.1089/wound.2013.0478. PMID: 24761333.

Qingxiu Dong, Lei Li, Damai Dai, Ce Zheng, Zhiyong Wu, Baobao Chang, Xu Sun, Jingjing Xu, and Zhifang Sui. A survey on in-context learning. *arXiv preprint arXiv:2301.00234*, 2022.

Seppo Enarvi, Marilisa Amoia, Miguel Del-Agua Teba, Brian Delaney, Frank Diehl, Stefan Hahn, Kristina Harris, Liam McGrath, Yue Pan, Joel Pinto, et al. Generating medical reports from patient-doctor conversations using sequence-to-sequence models. In *Proceedings of the first workshop on natural language processing for medical conversations*, pages 22–30, 2020.

Gregory Finley, Wael Salloum, Najmeh Sadoughi, Erik Edwards, Amanda Robinson, Nico Axtmann, Michael Brenndoerfer, Mark Miller, and David Suendermann-Oeft. From dictations to clinical reports using machine translation. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 3 (Industry Papers)*, pages 121–128, 2018.

George Hripcsak, David K Vawdrey, Matthew R Fred, and Susan B Bostwick. Use of electronic clinical documentation: time spent and team interactions. *Journal of the American Medical Informatics Association*, 18(2):112–117, 2011.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.

Aobo Kong, Shiwan Zhao, Hao Chen, Qicheng Li, Yong Qin, Ruiqi Sun, Xin Zhou, Enzhi Wang, and Xiaohang Dong. Better zero-shot reasoning with role-play prompting. *arXiv preprint arXiv:2308.07702*, 2023.

Alex Papadopoulos Korfiatis, Francesco Moramarco, Radmila Sarac, and Aleksandar Savkov. PriMock57: A dataset of primary care mock consultations. *arXiv preprint arXiv:2204.00333*, 2022.

Kundan Krishna, Sopan Khosla, Jeffrey P Bigham, and Zachary C Lipton. Generating SOAP notes from doctor-patient conversations using modular summarization techniques. *arXiv preprint arXiv:2005.01795*, 2020.

Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pages 74–81, 2004.

Lei Liu, Xiaoyan Yang, Fangzhou Li, Chenfei Chi, Yue Shen, Shiwei Lyu, Ming Zhang, Xiaowei Ma, Xiangguo Lv, Liya Ma, et al. Towards Automatic Evaluation for LLMs' Clinical Capabilities: Metric, Data, and Algorithm. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 5466–5475, 2024.

Alexander Mathioudakis, Ilona Rousalova, Ane Aamli Gagnat, Neil Saad, and Georgia Hardavella. How to keep good clinical records. *Breathe*, 12(4):369–373, 2016.

Genevieve B Melton, Clement J McDonald, Paul C Tang, and George Hripcsak. Electronic health records. In *Biomedical Informatics: Computer Applications in Health Care and Biomedicine*, pages 467–509. Springer, 2021.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pages 311–318, 2002.

Dave Van Veen, Cara Van Uden, Louis Blankemeier, Jean-Benoit Delbrouck, Asad Aali, Christian Bluethgen, Anuj Pareek, Malgorzata Polacin, Eduardo Pontes Reis, Anna Seehofnerová, et al. Adapted large language models can outperform medical experts in clinical text summarization. *Nature medicine*, 30(4):1134–1142, 2024.

Vivek Podder and Valerie Lew and Sassan Ghassemzadeh. *SOAP Notes*, book chapter SOAP Notes. StatPearls Publishing, Treasure Island (FL), January 2024. doi: NBK482263. URL https://www.ncbi.nlm.nih.gov/books/NBK482263/. Available from StatPearls [Internet], PMID: 29489268, Copyright © 2024, StatPearls Publishing LLC.

Junda Wang, Zonghai Yao, Zhichao Yang, Huixue Zhou, Rumeng Li, Xun Wang, Yucheng Xu, and Hong Yu. NoteChat: a dataset of synthetic doctor-patient conversations conditioned on clinical notes. *arXiv preprint arXiv:2310.15959*, 2023.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.

Ehsan Yaghmaei, Albert Pierce, Hongxia Lu, Yesha M Patel, Louis Ehwerhemuepha, Ahmad Rezaie, Seyed Ahmad Sajjadi, and Cyril Rakovski. A causal inference study: The impact of the combined administration of Donepezil and Memantine on decreasing hospital and emergency department visits of Alzheimer's disease patients. *PLoS One*, 18(9):e0291362, 2023.

Wen-wai Yim and Meliha Yetisgen-Yildiz. Towards automating medical scribing: Clinic visit dialogue2note sentence alignment and snippet summarization. In *Proceedings of the Second Workshop on Natural Language Processing for Medical Conversations*, pages 10–20, 2021.

Wen-wai Yim, Asma Ben Abacha, Griffin Adams, Neal Snider, and Meliha Yetisgen. Overview of the MEDIQA-Sum Task at ImageCLEF 2023: Summarization and Classification of Doctor-Patient Conversations. In *CLEF (Working Notes)*, pages 1347–1360, 2023a.

Wen-wai Yim, Yujuan Fu, Asma Ben Abacha, Neal Snider, Thomas Lin, and Meliha Yetisgen. Acibench: a novel ambient clinical intelligence dataset for benchmarking automatic visit note generation. *Scientific Data*, 10(1):586, 2023b.

Longxiang Zhang, Renato Negrinho, Arindam Ghosh, Vasudevan Jagannathan, Hamid Reza Hassanzadeh, Thomas Schaaf, and Matthew R Gormley. Leveraging pretrained models for automatic summarization of doctor-patient conversations. *arXiv preprint arXiv:2109.12174*, 2021.

Zhengyun Zhao, Qiao Jin, Fangyuan Chen, Tuorui Peng, and Sheng Yu. Pmc-patients: A large-scale dataset of patient summaries and relations for benchmarking retrieval-based clinical decision support systems. *arXiv preprint arXiv:2202.13876*, 2022.

Jianwei Zheng, Islam Abudayyeh, Cyril Rakovski, Louis Ehwerhemuepha, Ahmad Rezaie Mianroodi, Jay N Patel, Alomari Ihab, and Chizobam Ani. Trends in heart failure costs for commercially insured patients in the United States (2006–2021). *BMC Health Services Research*, 24(1):780, 2024.

Appendix A. Prompts

This appendix contains the prompts we used in our experiments.

Assume you are a very experienced physician and you are conducting research. The research project is to generate synthetic medical notes from doctor-patient conversations. Your job is to provide a scenario for the note to be generated.

The notes must contain these variables:

** 1) Medical Outcome: Like diagnosis, prescribed treatment, follow-up recommendations, referral to specialists, referral to further tests or imaging, medication adjustment, lifestyle change (sleep, diet, exercise, tobacco use, alcohol use). If you think the note should contain prescribing medication, details must be included like dose, units, frequency, duration, quantity, quantity type (like tablets, etc.), and route (like oral or injected). If you think the note should contain referral, it must include details like the reason for the referral, the specialty, and the doctor's name. If you think the note should contain an order for blood work, it must include details like if it is for biochemistry, hematology, immunology, microbiology, viral hepatitis, vitamin D, prostate-specific antigen, or anything else that suits the scenario. You need to be very specific. If you think the note should contain an order for imaging, it must include details like the modality of the imaging and the area of the body. For example, if it is ultrasound, it can be an order for Abdominal, Thyroid, Musculoskeletal, Sonohysterogram, Sonohysterogram, Biophysical Profile(BPP), Scrotal, G.U. Tract - Kidneys-Bladder(Prostate), or anything else as it suits the scenario. Try to be very specific. REMEMEBER, you do not have to use all of them in the scenario. Use the ones that you see fit for the scenario.

** 2) Medical History: Can contain Previous Diagnoses, Family Medical History, Mdication History, Allergies, and Chronic Conditions.

** 3) Symptom Description: Can contain Severity, Duration, Associated Symptoms, Frequency, and Impact on Daily Activities.

** 4) Patient's self-reported habits and lifestyle: Can contain Sleep, Diet, Exercise, Tobacco Use, Alcohol Consumption, Drug Use, Recreational Activities.

** 5) Demographic Information: Can contain Age, Gender, Ethnicity, Socioeconomic Status, Education Level, Health Literacy, and Job Status.

** 6) Patient's Behavior: The level of patient's cooperation with medical advice.

** 7) Geographical Location: Can contain Big City vs Small City, Rural vs Urban, Pollution and Environmental Health Risks, Neighborhood Type- eg if Impoverished or Affluent, Well-served by Transit, Food Desert, etc.

** 8) Clinical Setting: Can contain Hospitals, Clinics, Telemedicine, Mommunity Health Services, Urgent Care Centers, Research Facilities, School Health Services, Private Practice, and Specialty Clinics.

** 9) Type of Encounter: Can contain Initial Consultation, Follow-up, Emergency Visit, Routine Check-up, Chronic Disease Management (regular appointments to manage long-term health conditions like diabetes, heart disease, or chronic pain), Preventive Health Screening.

** 10) Treatment Disparities: There may be a tendency to offer less aggressive treatment or fewer options due to assumptions about compliance or ability to pay.

** 11) Native or Non-Native English Speaking Patient.

** 12) Physical exams: Any physical exams that are suitable for the scenario and the disease. If nothing is suitable, use NA.

** 13) Investigation/Test results: Like any tests that have been done for the patient while visiting. The results could be ready and reviewed in the scenario or could be awaiting. If awaiting, you need to be very specific about what type of tests have been done. For example, if it is X-ray, you need to incude the details mentioned abve about the imaging. You can also use NA if you think tests are not suitable for the scenario.

The user will give you the ICD-10 description of the disease. The diagnosis in the scenario must be the ICD-10 description. First, you select a role for yourself. You can be a Family Medicine Physician, a General physician, or a specialist with different specialties.

Select the role based on the ICD-10 description and output it with the keyword 'ROLE:'. Second, you must come up with a scenario and list all the values for the variables you want to use in the scenario. Do not output any extra text, just your role at the top of the scenario and the list of the values. You should incorporate medication and blood work or imaging requests in the scenarios with the details mentioned above if it suits the scenario. These are artificial and people will not be using it without asking a real doctor.

The user will evaluate the scenario you provided. If they accept it, they do not give any feedback. If they do not accept your scenario, they will give you feedback about how to improve the scenario and you must incorporate the feedback and generate a new scenario.

Below is an example of a medical note. Remember, you need to provide the scenario, not the note.

EXAMPLE NOTE

Figure 2: Prompt for Scenario Provider LLM in Note Generation Pipeline

---

Assume you are a very experienced physician and you are conducting research. The research project is to generate synthetic medical notes from doctor-patient conversations. You have a coworker that works with you on the project. You have different roles. The coworker provides you with the scenario they are going to write notes with. Your job is to judge whether the scenario is approved or not based on the conditions I provided below.

The scenarios have 13 variables.

** 1) Medical Outcome

** 2) Medical History

** 3) Symptom Description

** 4) Patient's self-reported habits and lifestyle

** 5) Demographic Information

** 6) Patient's Behavior

** 7) Geographical Location

** 8) Clinical Setting

** 9) Type of Encounter

** 10) Treatment Disparities

** 11) Native or Non-Native English Speaking Patient.

** 12) Physical exams: Any physical exams that are suitable for the scenario.

** 13) Investigation/Test results: Like any tests that have been done for the patient while visiting. The results could be ready and reviewed in the scenario or could be awaiting.

You have to check three conditions and then decide to approve or deny the scenario:

** a) A pair-wise comparison of the values of the variables. You need to check if at least 4 out of 13 of the values of the variables in the scenario are different from the previously approved scenarios.

** b) You need to check if the scenario is medically correct in terms of symptoms, tests, diagnosis, and treatment.

** c) You need to check if the scenario is plausible or not.

First, check condition (a). If it is not met, the scenario is rejected and you do not need to check conditions (b) and (c). If the scenario passes these three conditions, then say "Go". If not, you say "NoGo". In the case that there is no scenario previously approved, you should only check conditions (b) and (c). If your decision is "Go", you must only return "Go". Else, output your reasons and provide feedback to help your coworker about what they can do to generate a scenario to pass the above three conditions.

REMEMEBER: IF YOU APPROVE THE SCENARIO, YOU MUST ONLY OUTPUT LIKE BELOW. YOU CANNOT USE ANY BOLD OR ITALIC OR HEADING OR ANYTHING ELSE: DECISION: Go

---

Figure 3: Prompt for Scenario Judge LLM in Note Generation Pipeline

---

Assume you are a very experienced physician and you are conducting research.

The research project is to generate synthetic medical notes from doctor-patient conversations. The notes must be in this format:

** 1. Subjective: This section includes the patient's own description of their symptoms and complaints. Roll a dice, if the result is odd, break this part down into several sub-parts like Chief Complaint (CC), History of Present Illness (HPI), Review of Systems (ROS).

** 2. Objective: This section includes observations and data gathered by the physician, such as vital signs, physical examination findings, and test results.

** 3. Assessment: This section includes the physician's evaluation of the patient's condition, including a diagnosis or differential diagnosis.

** 4. Plan: This section includes the physician's recommendations for treatment, management, and follow-up.

You will be given a scenario containing your role. Your role can be a Family Medicine Physician, a General physician, or a specialist with different specialties. Your task is to generate the note based on the scenario. The note you generate must be in the format mentioned above.

Below is an example of a high quality medical note:

EXAMPLE NOTE

Figure 4: Prompt for Note Writer LLM in Note Generation Pipeline

Assume you are a very experienced physician and you are conducting research. The research project is to generate synthetic medical notes from doctor-patient conversations. The notes must be in this format:
** 1. Subjective: This section includes the patient's own description of their symptoms and complaints.
** 2. Objective: This section includes observations and data gathered by the physician, such as vital signs, physical examination findings, and test results.
** 3. Assessment: This section includes the physician's evaluation of the patient's condition, including a diagnosis or differential diagnosis.
** 4. Plan: This section includes the physician's recommendations for treatment, management, and follow-up.
You will be given a note. Your task is to polish the note and make sure the information is placed correctly in the relevant section. You cannot add or remove any information, except where you have been given permission. Make sure of these:
** a) If the doctor is ordering an imaging or bloodwork to be done, it must come under the "Plan" section. But if it is already done, it must come under the "Objective" section.
** b) If the doctor is prescribing a medication or renewing a medication, changing doses, etc., it must be under the "Plan" section.
** c) If the patient is being refered to another doctor, the referrals must come under the "Plan" section. The referral must contain:
1) The reason for referral
2) The specialty of the doctor
3) The doctor's name
If any of the three parts mentioned above about the refereal is missing, add it. If you need add a name for the doctor, choose an appropriate name, be creative and realistic in choosing the names.
** d) Patients' must have names. If there is no name, add it. Choose an appropriate name, be creative and realistic in choosing the names.
** e) The output must only contain the medical note. If there is anything extra at the beginning or at the end, you should remove it. For example, there could be thinng like "note is: Medical Note" at the beggining or things like " Note: Please ensure this note is reviewed by the attending healthcare provider or physician for accuracy and completeness before being added to the patient's medical record." at the end of the medical note. Please remove them so that the output is only the medical note itself, not anything else. Just output the revised note, not anything else.

Figure 5: Prompt for Note Polisher in Note Generation Pipeline

You are a helpful medical research assistance. You will be give a medical note and your task is to generate the conversation between the doctor and the patient that led to that note.
Your conversation must include all information. if it's difficult to include them all, you can use the original sentences in the notes. The common symptoms and common medical history should be told by patient.
Some specific symptoms and medical history should be added by the doctor after the patient has finished describing his symptoms and medical history.
For example:
Doctor: Can you give me your medical history record?
Patient: Here you are.
Doctor: Based on your medical history record...
Because after patient has finished describing common symptoms or medical history, he will give doctor his medical history records.
After patient give the doctor his medical history record, the doctor could know medical history record. Otherwise he didn't know any information of the medical history.
Some result should not come from history clinical note they should come from examination.
All the examination result, history examination result, vital signs and medical number must be told by doctor.
You could expand the parts of doctor to include more key words. If it is difficult to include you could just use the sentence of clinical note. The revised conversation should be at least around 80 to 150 utterances(doctor or patient should not say too much information at once).
The conversation must include all the information of the clinical note. You must include all the key words I gave you. If it is difficult to include all the key words you could use original the sentences of clinical note.
You cannot revise or eliminate any key words and you cannot use synonyms of the key words.
You shoudn't use the abbreviation if you know the full name(you should use full name not abbreviation, such as D9 must be day 9, D7 must be day 7. If both the full name and the abbreviation appear, it's better to use the full name rather than the abbreviation.
Patients must not say any highly specialized terms, medical terminology or medical dosage. They can only describe limited common symptoms. The doctor should supplement the remaining information based on test results.
Don't repeat the same information in long paragraphs. The utterance of the dialogue needs to be expanded as much as possible. The patient and the doctor should have many modal particles (e.g. hmm, yes, okay) to increase interaction. Pay attention to the examples below and try to incorporate non-linear discussions to make it more realistic.
You cannot use[Patient's Name] or any other plcae holder in the dialogue.
Here are a good real note and dialogue example:
Example 1:
Note:
EXAMPLE 1 NOTE
conversations:
EXAMPLE 1 DIALOGUE
<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<
Example 2:
Note:
EXAMPLE 2 NOTE
conversations:
EXAMPLE 2 DIALOGUE
<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<
Example 3:
Note:
EXAMPLE 3 NOTE
conversations:
EXAMPLE 3 DIALOGUE
<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<
You must follow the structure of the dialogues in the examples above.
The number of utterance should be at least 80 and sometimes patient didn't clearly hear and he could say parden to let the doctor say again. The dialogue must be in English. Your job is to only generate the dialogue. You cannot generate summary notes.

Figure 6: Prompt for Dialogue Generator in Dialogue Generation Pipeline

Expand the conversation. You must add chit chats to the conversation. The conversation for patient parts can be more colloquial. The patient and the doctor should have many modal particles (e.g. hmm, yes, okay) to increase interaction.
All the numbers and medical concepts that appear in the note should be mentioned by the doctor.
Professional medical terms and numbers should always occur in the doctor's utterances but not in the patient's answer. The doctor may describe and explain professional judgment to the patient and instruct the patient on follow-up requirements, but not ask questions that require professional medical knowledge to answer. All the information of medical history, symptoms and medication history should be told by patient.
The patient's answer should be succinct and accurate in a colloquial lay language style. The answer must align with the clinical notes and as colloquial as possible.
You can add some transitional phrases to make the conversation more logical. For example:
Example 1:
Patient: I understand, please go ahead.
(After examination)
Doctor: The result shows......
Your conversations can follow the logical sequence of a doctor's inquiry.
The conversations must be coherent and cohesive. For example, the output cannot be seperated by texts like "HISTORY OF PRESENT ILLNESS" or "SOCIAL HISTORY".
Extra information that does not fit into the conversation should not be added to the output. For example, below is an extra information that should be removed from the output:
<<<<<<<>>>>>>>>
- **INSTRUCTIONS**
Patient Agreements: The patient understands and agrees with the recommended medical treatment plan.
<<<<<<<<>>>>>>>>
Patients should not say too much information at once.
ICD code of the disease must not be present in the dialogue. If it is present, remove it.
There should not be any extra information at the beggining or at the end of the conversation. For example, "dialogue is:" should not be present in the output. You must make sure you only return the dialogue itself, not anything extra. You cannot add phrases like "dialogue is: Certainly! Let's expand the conversation with more colloquial language for the patient and professional details for the doctor".
If there are only the doctor and the patient present in the dialogue, the utterances must follow these indicators:
[doctor]: ...
[patient]: ...
If there are more people present in the dialogue, make sure to include all of them with a seperate indicator. For example, if the mother of the patient is present in the dialogue, use this indicators:
[doctor]: ...
[mother]: ...
[patient]: ...
All the information in the dialogue must align with the medical note below:
<<<<<<<>>>>>>>>
MEDICAL NOTE
<<<<<<<>>>>>>>>

Figure 7: Prompt for Dialogue Poliser in Dialogue Generation Pipeline

You are an assistant for medical professionals, specializing in summarizing their conversations with patients. Your role is to accurately and comprehensively summarize these conversations in the SOAP (Subjective, Objective, Assessment, Plan) format. Ensure that each summary is thorough and precise, reflecting all relevant details from the conversation to provide a reliable medical record.

Figure 8: Prompt for Dialogue-to-Note Task in Evaluation Pipeline

You are an assistant for medical professionals, specializing in generating synthetic doctor-patient dialogues based on medical notes. Your role is to create dialogues that realistically reflect potential interactions between doctors and patients. The notes will be given to you in the SOAP (Subjective, Objective, Assessment, Plan) format. Ensure that each dialogue is realistic and informative, encapsulating all relevant details from the medical notes to simulate an authentic conversation.

Figure 9: Prompt for Note-to-Dialogue in Evaluation Pipeline

Appendix B. Hyperparameters and data distribution

This section contains details of hyperparameters in our experiments and the distribution of clinical note data lengths, by subtoken, for additional background.

Scenario Generator configuration in Note Generation Pipeline

```
scenario_generator_config = {
"model": "gpt-4o",
"temperature": 1,
"max_tokens": 4000,
"top_p": 1,
"frequency_penalty": 0,
"presence_penalty": 0,
}
```

Figure 10: Configuration for Scenario Generator in the Note Generation Pipeline

Scenario Judge configuration in the Note Generation Pipeline

```
scenario_judge_config = {
"model": "gpt-4o",
"temperature": 0,
"max_tokens": 4000,
"top_p": 1,
"frequency_penalty": 0,
"presence_penalty": 0,
}
```

Figure 11: Configuration for Scenario Judge in the Note Generation Pipeline

Note Generator configuration in the Note Generation Pipeline

```
note_generator_config = {
"model": "gpt-4o",
"temperature": 0.9,
"max_tokens": 4000,
"top_p": 1,
"frequency_penalty": 0,
"presence_penalty": 0,
}
```

Figure 12: Configuration for Note Generator in the Note Generation Pipeline

Note Polisher Configuration in the Note Generation Pipeline

```
note_polisher_config = {
"model": "gpt-4o",
"temperature": 0,
"max_tokens": 4000,
"top_p": 1,
"frequency_penalty": 0,
"presence_penalty": 0,
}
```

Figure 13: Configuration for Note Polisher in the Note Generation Pipeline

Dialogue Generator Configuration in the Dialogue Generation Pipeline

```
dialogue_generator_config = {
"model": "gpt-4o",
"temperature": 0.7,
"max_tokens": 4095,
"top_p": 1,
"frequency_penalty": 0,
"presence_penalty": 0,
}
```

Figure 14: Configuration for Dialogue Generator in the Dialogue Generation Pipeline

Dialogue Polisher configuration in the Dialogue Generation Pipeline

```
dialogue_polisher_config = {
"model": "gpt-4o",
"temperature": 0.5,
"max_tokens": 4095,
"top_p": 1,
"frequency_penalty": 0,
"presence_penalty": 0,
}
```

Figure 15: Configuration for Dialogue Polisher in the Dialogue Generation Pipeline

Defining the configuration for the base model, LoRA and training

```
tuning_config = {
"model_config": {
"max_seq_length": 8192,
"dtype": torch.bfloat16,
"load_in_4bit": True,
},
"lora_config": {
"r": 16,
"target_modules": ["q_proj", "k_proj",
"v_proj", "o_proj", "gate_proj", "up_proj",
"down_proj"],
"lora_alpha": 16,
"lora_dropout": 0,
"bias": "none",
"use_gradient_checkpointing": True,
"use_rslora": False,
"use_dora": False,
"loftq_config": None,
},
"training_config": {
"per_device_train_batch_size": 2,
"gradient_accumulation_steps": 4,
"warmup_steps": 5,
"max_steps": 0,
"num_train_epochs": 4,
"learning_rate": 2e-4,
"fp16": not torch.cuda.is_bf16_supported(),
"bf16": torch.cuda.is_bf16_supported(),
"logging_steps": 1,
"optim": "adamw_8bit",
"weight_decay": 0.01,
"lr_scheduler_type": "linear",
"seed": 42,
"output_dir": "outputs"
}
}
```

Figure 16: Configuration for the base model, LoRA, and training setup in tuning models for evaluating the usefulness of MedSynth for the Dial-2-Note and Note-2-Dial tasks; Unsloth library was used to achieve higher speed

Updated Tuning Configuration

```
tuning_config = {
"model_config": {
"max_seq_length": 8192,
"dtype": torch.bfloat16,
"load_in_4bit": True,
},
"lora_config": {
"r": 16,
"target_modules": ["q_proj", "k_proj",
"v_proj", "o_proj", "gate_proj", "up_proj",
"down_proj"],
"lora_alpha": 16,
"lora_dropout": 0,
"bias": "none",
"use_gradient_checkpointing": True,
"use_rslora": False,
"use_dora": False,
"loftq_config": None,
},
"training_config": {
"per_device_train_batch_size": 2,
"gradient_accumulation_steps": 4,
"warmup_steps": 5,
"max_steps": 0,
"num_train_epochs": 1,
"learning_rate": 2e-4,
"fp16": not torch.cuda.is_bf16_supported(),
"bf16": torch.cuda.is_bf16_supported(),
"logging_steps": 1,
"optim": "adamw_8bit",
"weight_decay": 0.01,
"lr_scheduler_type": "linear",
"seed": 42,
"output_dir": "outputs"
}
}
```

Figure 17: Configuration for the base model and LoRA training for tuning the models used in benchmarking; Unsloth library was used to achieve higher speed

Model evaluator generation configuration

```
config = {
"max_new_tokens": 3000,
"do_sample": True,
"temperature": 0.6,
"top_p": 0.9,
"use_cache": True,
}
```

Figure 18: Configuration for inference of tuned models in tuning models for evaluating the usefulness of MedSynth for the Dial-2-Note and Note-2-Dial tasks; Unsloth library was used to achieve higher speed

GPT-4o model configuration used in benchmarking

```
config = {
"model": "gpt-4o",
"temperature": 0.5,
"max_tokens": 4000,
"top_p": 1,
}
```

Figure 19: GPT-4o model configuration used in benchmarking

Inference configuration of tuned models used in benchmarking

```
config = {
"max_new_tokens": 4000,
"do_sample": True,
"temperature": 0.5,
"top_p": 1,
"use_cache": True,
}
```

Figure 20: Inference configuration of tuned models used in benchmarking; Unsloth library was used to achieve higher speed

Appendix C. Details about Roles in Scenario Provider LLM

The analysis of the generated scenarios shows that there are 258 unique roles in the whole dataset. Family Medicine Physician is the most common role, followed by Orthopedic Specialist, Neurologist, and General Physician. Table 6 shows the distribution of roles, and Table 7 shows the top 20 roles.

Table 6: Frequency distribution of roles in Scenario Provider LLM

# of Occurrences	Count
More than 1000	1
Between 500 and 1000	4
Between 100 and 500	18
Between 50 and 100	6
Between 10 and 50	26
Below 10	203

Table 7: Top 20 most frequent roles in Scenario Provider LLM

Role	Frequency
Family Medicine Physician	1960
Orthopedic Specialist	613
Neurologist	584
General Physician	556
Gastroenterologist	526
Cardiologist	490
Dermatologist	371
Pulmonologist	326
Oncologist	326
Endocrinologist	305
Ophthalmologist	296
Obstetrician	289
Urologist	283
Orthopedic Surgeon	233
General Surgeon	221
Gynecologist	196
Psychiatrist	193
Nephrologist	166
Hematologist	159
Rheumatologist	153

Appendix D. Comparison of Notes Generated by Models Trained on MedSynth and NoteChat

When tuned solely on NoteChat, the model failed to adhere to the SOAP structure in generating medical notes from dialogues. In the same setting, the model tuned on MedSynth successfully generated notes in SOAP format. This appendix shows an example of such a situation.

CHIEF COMPLAINT
Rheumatoid arthritis management.
HISTORY OF PRESENT ILLNESS
Mr. Nelson is a 72-year-old male who presents today for ongoing management of rheumatoid arthritis. He was last seen by me on 03/04/2021, where he received a steroid injection to his right knee. He continues to experience right knee pain and swelling. The pain level is 6/10. He notes when he stands after he has been sitting for a while, he has to use a cane to "get his bearings." The injection he received to the right knee at the last visit was only beneficial for a few weeks. The patient notes whole right leg swelling since the last visit. He usually wears support hose, which helps to some degree. He has not attended physical therapy.
PHYSICAL EXAM
Musculoskeletal Lower right extremity: Swollen right knee, slightly reduced range of motion. Skin warm to touch. Lower right leg pits with pressure.
PROCEDURE
Fluid aspiration and Steroid injection. Risks, benefits, and alternatives were explained. The risks of steroid injection were explained, including but not limited to, hypopigmentation, soft tissue atrophy, tendon ruptures, and transient hyperglycemia among others. Verbal consent was obtained. The right knee was accessed through the lateral approach with 3 mL of initially clear fluid then blood-tinged fluid. Then the right knee was prepped in a standard fashion. The right knee was injected with 80 mg triamcinolone and 1 mL lidocaine. The procedure was tolerated well with no complications. Post procedure instructions were given.
ASSESSMENT
• Rheumatoid arthritis management. • High risk medication use. • Right knee pain. • Lower extremity pitting edema.
PLAN
Rheumatoid arthritis management. The patient will continue Xeljanz XR 11 mg daily. He will also continue Celebrex 200 mg daily.
High-risk medication use. The patient is currently on the high-risk medication Xeljanz. We will check his CMP levels and monitor periodically every couple of months.
Right knee pain. He has right knee pain and swelling, with pain level 6/10. On exam today, the right knee continues to be hot, which is ongoing inflammation. We performed a fluid aspiration of the right knee today, see procedure note above. A steroid injection to the right knee was given today, see procedure note above. We will arrange for x-ray of the right knee and refer his to physical therapy.
Lower extremity pitting edema. The patient notes whole right leg swelling since the last visit. He usually wears support hose, which helps to some degree. He has not seen his PCP. We discussed compression stockings, elevate feet, and a low salt diet. He will check with his PCP for further instructions.

Figure 21: Ground Truth Note

Tyler Nelson, a 70-year-old male with a date of birth of 3/6/1949, social security number 385-729- and a home address of [redacted] presented to the office for ongoing management of rheumatoid arthritis. He received a steroid injection in the right knee last time he was in the office. He stated that the injection helped for a couple of weeks, it's just it. He stated that his right knee is swollen, and it's painful. He stated that his pain is a 3 or so on the pain scale, typically on a normal day it's around a 6. It's been swollen like that for how long now? He stated that it's been about two months now. Before it was just the knee was not swollen like this, but he stated that he tried to wear compression socks a lot of the time of the leg, but it has not been helping. He stated that he slowed down a little bit, and he found a few things too fast. When it really hurts is when he's been sitting for a while, and then he's got to try and get up. He's so stiff getting up he's got to use a cane to get his bearings. He means that after a little bit, it loosens up, but that first getting up is real difficult. He is on xeljanz XR 11 milligrams a day, he's taking it everyday, and he's also taking celebrex which he started maybe 2 months ago. The patient sat down and removed his socks and shoes. The right leg was swollen and hot. The patient stated that it hurts when he tries to straighten it all the way. Range of motion slightly diminished. The patient stated that the whole thing is swollen, not just the knee. There was some pitting around the knee and ankle. Normal capillary refill in the toes. Does any of this area hurt? The patient stated that it's kinda sore, like an old bruise? The patient agreed to get an x-ray of the knee. The patient was called for physical therapy a couple of years ago for the other knee. So, it was set up for him. The patient stated that it's been helpful to some degree, but it's still swollen, especially around the knee. The patient stated that he's got to try and get up, and it's real difficult. The patient was told that we'll set that up for him now. We'll also do an x-ray on that knee. We'll also set him for a physical therapy, and if nothing is helping then we really might need to get a surgery consult to see what we can do with that knee. The patient agreed to go ahead and we gave him the steroid shot. We tried to get a little bit of fluid out before we gave him the injection, there was no fluid but we saw if we could get anything out regardless. The procedure was explained, the risks, benefits, and alternatives were explained. The risks of steroid injection were explained, including, but not limited to: hypopigmentation, soft tissue atrophy, tendon ruptures, and transient hypoglycemia among others. Verbal consent was attained. The right knee was, uh, um, accessed through the lateral approach with 3 milliliters of initially clear fluid, then bloo- blood tinged fluid. Then the right knee was prepped in a standard-fashion. The right knee was injected with 80 milligrams of triamcinolone, and one milliliter of lidocaine. The procedure was tolerated well, with no complications. Post-procedure instructions were given. Patient will follow up with his pcp. Thank you.

Figure 22: Note Generated by the Model Trained Solely on NoteChat

1. Subjective:
Chief Complaint (CC): Swelling and pain in the right knee.
History of Present Illness (HPI): Tyler Nelson is a 73-year-old male with a history of rheumatoid arthritis who presents for a follow-up visit regarding the management of his condition. He reports that he experienced significant swelling in his right knee following a recent steroid injection administered during his last visit. The swelling has persisted despite his use of compression socks, which he has been wearing frequently to alleviate symptoms. The pain is moderate, with a pain score of 6/10 on a normal day and currently 3/10, and has been present for the past 2 months. He notes that the swelling and pain have limited his mobility, particularly when getting up from sitting, where he requires the use of a cane.
He has been compliant with his medication regimen, including Xeljanz 11mg daily and Celebrex 200mg daily, which he started 2 months ago. He is motivated to follow medical advice and is highly cooperative.
Review of Systems (ROS): - **Musculoskeletal:** Swelling and pain in the right knee, moderate severity, limiting mobility. - **General:** No recent weight loss, fever, or chills. - **Cardiovascular:** No chest pain or palpitations. - **Gastrointestinal:** No nausea, vomiting, or abdominal pain. - **Neurological:** No headaches or dizziness. - **Dermatological:** No rashes or skin changes.
2. Objective:
Physical Examination: - **Vital Signs:** Stable - **Musculoskeletal:** - Right knee: Swelling, erythema, and pitting noted. Range of motion slightly diminished. Increased heat around the knee. Capillary refill within normal limits in the toes. - **Cardiovascular:** Regular rate and rhythm, no murmurs. - **Respiratory:** Clear to auscultation bilaterally. - **Neurological:** Alert and oriented, no focal deficits.
Results: - **Blood Work:** CMP within normal limits. Xeljanz trough levels checked; levels are within the therapeutic range.
3. Assessment:
1. **Rheumatoid Arthritis with right knee effusion, not elsewhere classified.** - **Medical Reasoning:** The patient presents with persistent swelling and moderate pain in the right knee, consistent with effusion. The swelling has limited his mobility and daily activities. - **Differential Diagnosis:** Septic arthritis, crystal-induced arthritis, gout, lymphedema.
4. Plan:
1. **Treatment:** - Administered steroid injection (Triamcinolone 80mg, 1mL) to the right knee today. - Fluid aspiration performed during the procedure; 3mL of initially clear fluid, then blood-tinged fluid was aspirated.
2. **Imaging:** - Order an X-ray of the right knee to evaluate for possible underlying causes of effusion.
3. **Physical Therapy:** - Referral for physical therapy to aid in improving range of motion and strength.
4. **Follow-up Recommendations:** - Follow-up visit in 2 weeks for re-evaluation of the knee. - Referral to Dr. Emily Carter, a rheumatologist, for further management of rheumatoid arthritis.
5. **Patient Education and Counseling:** - Discussed the importance of reducing salt intake to help manage edema. - Encouraged the patient to elevate the legs when possible to reduce swelling. - Advised the patient to continue using compression socks as a preventive measure. - Discussed the potential need for surgical intervention if conservative measures fail to improve symptoms.
6. **Medication Adjustment:** - Continue Xeljanz 11mg daily. - Continue Celebrex 200mg daily. - Regular monitoring of Xeljanz trough levels every 2 months due to the high-risk medication profile.
Patient Agreements: The patient understands and agrees with the recommended medical treatment plan.
INSTRUCTIONS: The patient will follow up in 2 weeks for re-evaluation and will have an X-ray of the right knee performed. He will also make an appointment with Dr. Emily Carter, a rheumatologist, and ensure regular blood work for Xeljanz trough levels.

Figure 23: Note Generated by the Model Trained Solely on MedSynth

Appendix E. Figures

Figure 24 shows the results of our survey on the important variables in determining the quality of medical notes. Figure 25 demonstrates the reason why embedding-based metrics cannot be used in our study to evaluate performance. Our notes have a larger token length than the models underlying these metrics, which typically have a context length of 512 tokens.

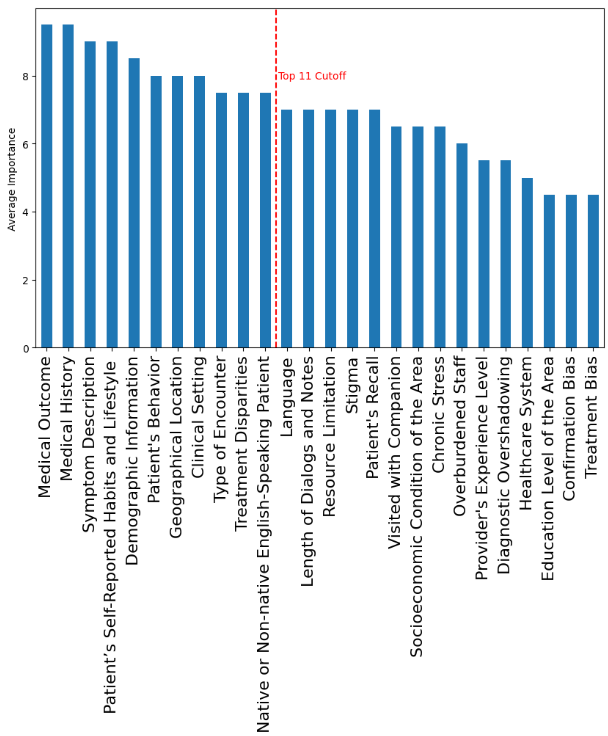

Figure 24: Results of Important Variables Survey

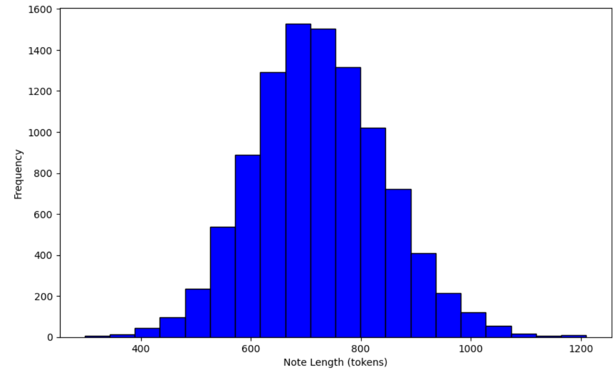

Figure 25: BERT Subtoken Lengths of Notes; The length of our notes are longer than context length of embedding-based performance evaluators

Appendix F. Data Statistics Tables

This appendix contains several statistics related to MedSynth and IQVIA. Table 8 shows the comparison of MedSynth with Aci-Bench and real data, while Table 9 indicates the statistics of MedSynth's train and test sets used for benchmarking. In addition, Table 10 shows the top most prevalent ICD-10 codes in our analysis.

Table 8: Comparison of alignment statistics for 20 samples across MedSynth, Aci-Bench, and a real doctor-patient dataset. For consistency with real-world data, the comparison utilized data from Table 5 of the Aci-Bench study, which provided alignment statistics for 20 samples.

	MedSynth	**Aci-Bench**	**Real Data**
dialogue			
avg length (no speaker tokens) (tok)	970	1505	1203
avg length (sentences)	57	141	80
note			
avg length (tok)	618	683	492
avg length (sentences)	23	66	49

Table 9: Statistics of the train and test sets for benchmarking

	Train	**Test**
# of data points	8529	1506
# of unique ICD-10s	2001	1131
dialogue		
avg length (no speaker tokens) (tok)	931	938
avg length (sentences)	55	56
note		
avg length (tok)	621	623
avg length (sentences)	23	23

Table 10: Top 20 most frequent ICD-10 codes in IQVIA

ICD-10 Description	**Frequency**
ESSENTIAL (PRIMARY) HYPERTENSION	21,788,529
ENCOUNTER FOR GENERAL ADULT MEDICAL EXAMINATION WITHOUT ABNORMAL FINDINGS	19,497,471
ENCOUNTER FOR ROUTINE CHILD HEALTH EXAMINATION WITHOUT ABNORMAL FINDINGS	13,272,519
ENCOUNTER FOR IMMUNIZATION	10,470,154
TYPE 2 DIABETES MELLITUS WITHOUT COMPLICATIONS	9,890,112
END STAGE RENAL DISEASE	9,040,289
HYPERLIPIDEMIA, UNSPECIFIED	8,061,616
LOW BACK PAIN	7,336,799
OBSTRUCTIVE SLEEP APNEA (ADULT) (PEDIATRIC)	7,073,161
ILLNESS, UNSPECIFIED	6,253,356
HYPOTHYROIDISM, UNSPECIFIED	5,618,816
ENCOUNTER FOR SCREENING MAMMOGRAM FOR MALIGNANT NEOPLASM OF BREAST	5,393,554
ENCOUNTER FOR GYNECOLOGICAL EXAMINATION (GENERAL) (ROUTINE) WITHOUT ABNORMAL FINDINGS	5,299,912
URINARY TRACT INFECTION, SITE NOT SPECIFIED	5,075,284
OTHER LONG TERM (CURRENT) DRUG THERAPY	4,724,627
CERVICALGIA	4,621,704
ATHEROSCLEROTIC HEART DISEASE OF NATIVE CORONARY ARTERY WITHOUT ANGINA PECTORIS	4,504,152
CHEST PAIN, UNSPECIFIED	4,396,428
GASTRO-ESOPHAGEAL REFLUX DISEASE WITHOUT ESOPHAGITIS	4,224,803
ACUTE KIDNEY FAILURE, UNSPECIFIED	3,774,237